Table 1: Performance on ALFWorld and WebShop. Results are averaged over 3 random seeds. For ALFWorld, we report the average success rate (%) for each subtask as well as the overall result. For WebShop, we report both the average score and the average success rate (%). GiGPO$_{\text{w/ std}}$ denotes using $F_{\text{norm}} = \text{std}$, while GiGPO$_{\text{w/o std}}$ uses $F_{\text{norm}} = 1$.

Type	Method	ALFWorld							WebShop	
		Pick	Look	Clean	Heat	Cool	Pick2	All	Score	Succ.
Base: Closed-Source Model										
Prompting	GPT-4o	75.3	60.8	31.2	56.7	21.6	49.8	48.0	31.8	23.7
Prompting	Gemini-2.5-Pro	92.8	63.3	62.1	69.0	26.6	58.7	60.3	42.5	35.9
Base: Qwen2.5-1.5B-Instruct										
Prompting	Qwen2.5	5.9	5.5	3.3	9.7	4.2	0.0	4.1	23.1	5.2
Prompting	ReAct	17.4	20.5	15.7	6.2	7.7	2.0	12.8	40.1	11.3
Prompting	Reflexion	35.3	22.2	21.7	13.6	19.4	3.7	21.8	55.8	21.9
RL Training	PPO (with critic)	$64.8_{\pm 3.5}$	$40.5_{\pm 6.9}$	$57.1_{\pm 4.9}$	$60.6_{\pm 6.6}$	$46.4_{\pm 4.0}$	$47.4_{\pm 1.9}$	$54.4_{\pm 3.1}$	$73.8_{\pm 3.0}$	$51.5_{\pm 2.9}$
RL Training	RLOO	$88.3_{\pm 3.0}$	$52.8_{\pm 8.6}$	$71.0_{\pm 5.9}$	$62.8_{\pm 8.7}$	$66.4_{\pm 5.5}$	$56.9_{\pm 4.7}$	$69.7_{\pm 2.5}$	$73.9_{\pm 5.6}$	$52.1_{\pm 6.7}$
RL Training	GRPO	$85.3_{\pm 1.5}$	$53.7_{\pm 8.0}$	$84.5_{\pm 6.8}$	$78.2_{\pm 7.9}$	$59.7_{\pm 5.0}$	$53.5_{\pm 5.6}$	$72.8_{\pm 3.6}$	$75.8_{\pm 3.5}$	$56.8_{\pm 3.8}$
RL Training	**GiGPO$_{\text{w/ std}}$**	$94.4_{\pm 5.9}$	$67.5_{\pm 4.6}$	$\mathbf{94.8}_{\pm 3.8}$	$\mathbf{94.4}_{\pm 7.8}$	$\mathbf{79.8}_{\pm 4.7}$	$76.4_{\pm 5.4}$	$\mathbf{86.7}_{\pm 1.7}$	$83.1_{\pm 1.6}$	$65.0_{\pm 3.2}$
RL Training	**GiGPO$_{\text{w/o std}}$**	$\mathbf{96.0}_{\pm 1.4}$	$\mathbf{76.5}_{\pm 3.9}$	$91.8_{\pm 5.5}$	$91.3_{\pm 6.3}$	$71.7_{\pm 8.4}$	$\mathbf{79.5}_{\pm 7.7}$	$86.1_{\pm 4.7}$	$\mathbf{83.5}_{\pm 1.8}$	$\mathbf{67.4}_{\pm 4.5}$
Base: Qwen2.5-7B-Instruct										
Prompting	Qwen2.5	33.4	21.6	19.3	6.9	2.8	3.2	14.8	26.4	7.8
Prompting	ReAct	48.5	35.4	34.3	13.2	18.2	17.6	31.2	46.2	19.5
Prompting	Reflexion	62.0	41.6	44.9	30.9	36.3	23.8	42.7	58.1	28.8
RL Training	PPO (with critic)	$92.3_{\pm 4.0}$	$64.0_{\pm 8.4}$	$92.5_{\pm 2.4}$	$89.5_{\pm 7.0}$	$80.3_{\pm 2.0}$	$68.8_{\pm 8.3}$	$80.4_{\pm 2.7}$	$81.4_{\pm 3.1}$	$68.7_{\pm 5.1}$
RL Training	RLOO	$87.6_{\pm 4.3}$	$78.2_{\pm 8.3}$	$87.3_{\pm 5.8}$	$81.3_{\pm 7.6}$	$71.9_{\pm 5.2}$	$48.9_{\pm 8.4}$	$75.5_{\pm 4.6}$	$80.3_{\pm 3.2}$	$65.7_{\pm 4.0}$
RL Training	GRPO	$90.8_{\pm 5.1}$	$66.1_{\pm 6.7}$	$89.3_{\pm 5.4}$	$74.7_{\pm 6.9}$	$72.5_{\pm 5.4}$	$64.7_{\pm 7.3}$	$77.6_{\pm 5.2}$	$79.3_{\pm 2.8}$	$66.1_{\pm 3.7}$
RL Training	**GiGPO$_{\text{w/ std}}$**	$\mathbf{97.7}_{\pm 1.6}$	$82.7_{\pm 7.9}$	$\mathbf{98.8}_{\pm 1.6}$	$83.7_{\pm 7.2}$	$\mathbf{89.3}_{\pm 8.2}$	$79.2_{\pm 6.6}$	$\mathbf{90.8}_{\pm 1.3}$	$84.4_{\pm 2.9}$	$72.8_{\pm 3.2}$
RL Training	**GiGPO$_{\text{w/o std}}$**	$91.8_{\pm 5.4}$	$\mathbf{88.6}_{\pm 6.3}$	$95.9_{\pm 3.2}$	$\mathbf{90.2}_{\pm 2.6}$	$86.5_{\pm 5.5}$	$\mathbf{85.2}_{\pm 7.5}$	$90.2_{\pm 2.3}$	$\mathbf{86.2}_{\pm 2.6}$	$\mathbf{75.2}_{\pm 3.8}$

5.1 Experiment Setup

Environments. We train the LLM agents on two challenging benchmarks: ALFWorld [5] and WebShop [21]. *ALFWorld* is an embodied environment designed to assess the ability of LLM agents to perform multi-step decision-making. In each episode, the agent receives a text goal and must accomplish it through multi-turn interaction with the environment. It includes 4,639 task instances across six categories of common household activities: Pick & Place (Pick), Examine in Light (Look), Clean & Place (Clean), Heat & Place (Heat), Cool & Place (Cool), and Pick Two & Place (Pick2). *WebShop* is a complex, web-based interactive environment designed to test the LLM agents in realistic online shopping scenarios. To complete the task, the agent must interact with a simulated HTML-based shopping website to search for, navigate to, and ultimately purchase a suitable item. It contains over 1.1 million products and 12k user instructions, providing a rich and diverse action space.

Baselines. We compare our approach with a range of competitive baselines: (1) Closed-source LLMs: GPT-4o [1] and Gemini-2.5-Pro [2], which represent state-of-the-art capabilities in general-purpose reasoning and language understanding. (2) Prompting agents: ReAct [28] and Reflexion [29], which rely on in-context prompting to guide multi-step behavior without parameter updates. (3) RL training methods: PPO [41], a widely used actor-critic algorithm that requires an additional value model, and group-based critic-free methods RLOO [15, 16] and GRPO [17], which perform advantage estimation over trajectory groups.

Training details. We use Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct [3] as our base models. The weighting coefficient ω is set to 1 with no further tuning. All RL training methods (including ours and the baselines) use exactly the same hyperparameter configurations. The rollout group size N for group-based RL methods is set to 8. The LLM agent is prompted to first generate a chain-of-thought [61] enclosed within <think> </think> tags, followed by the action enclosed within <action> </action> tags. Full training settings and hyperparameter details are provided in Appendix D.1.

5.2 Performance

Table 1 demonstrates the strong performance of GiGPO across both ALFWorld and WebShop. As shown, Closed-source LLMs offer only moderate performance: Gemini-2.5-Pro reaches 60.3% success on ALFWorld and 35.9% on WebShop, while GPT-4o lags further behind. Open-source prompt-only agents (e.g., ReAct and Reflexion) show marginal improvements over vanilla prompting but still underperform, underscoring the difficulty of long-horizon control without post-training. RL

training brings substantial gains: PPO improves average ALFWorld success to 54.4% on the 1.5B model and 80.4% on the 7B model, with WebShop scores also increasing significantly. However, this comes at the expense of increased complexity: requiring a separate critic network, hyperparameter tuning, and longer training durations [62, 48]. GRPO and RLOO also yield strong performance while being more computationally efficient, demonstrating the effectiveness of group-based RL in large-scale LLM training. Nevertheless, their lack of fine-grained per-step feedback limits their ability to provide precise credit assignment across long horizons. In contrast, GiGPO overcomes this limitation with a two-level advantage estimation, enabling both GiGPO$_{\text{w/ std}}$ and GiGPO$_{\text{w/o std}}$ to consistently surpass GRPO and RLOO. In particular, GiGPO$_{\text{w/o std}}$ surpasses GRPO by 13.3% on ALFWorld and 10.6% on WebShop at 1.5B, and by 12.6% and 9.1%, respectively, at 7B. These results highlight GiGPO's superior ability to train LLM agents more effectively and more efficiently. We also find that GiGPO enables agents to exhibit emergent reasoning behavior (see Appendix E).

Lastly, we observe that the normalization factor F_{norm} is task-dependent rather than universally helpful. On relatively difficult tasks (such as Look, Pick2, and WebShop), standard-deviation scaling ($F_{\text{norm}} = \text{std}$) could exaggerate gradients from overly difficult samples or highly imbalanced groups, harming update stability; fixing $F_{\text{norm}} = 1$ therefore yields higher success. Yet, $F_{\text{norm}} = 1$ offers no clear advantage on other tasks and both variants perform similarly, which aligns with findings in [63]. This suggests that $F_{\text{norm}} = \text{std}$ can still be beneficial when reward variance is stable.

5.3 Ablation Study

Next, we conducted an ablation study, comparing GiGPO$_{\text{w/o std}}$ ($F_{\text{norm}} = 1$), GiGPO$_{\text{w/ std}}$ ($F_{\text{norm}} = \text{std}$), GiGPO$_{\text{w/o } A^S}$ (without step relative advantages), and GiGPO$_{\text{w/o } A^E}$ (without episode relative advantages) to evaluate the impact of each component on performance. We use Qwen2.5-1.5B-Instruct as the policy of the agent. The results are presented in Figure 4.

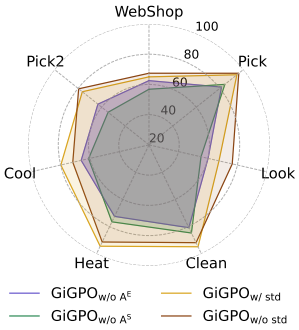

Figure 4: Ablation results. The y-axis shows success rate (%).

As illustrated, eliminating either component of the two-level advantage significantly degrades performance. Removing the episode relative advantages (GiGPO$_{\text{w/o } A^E}$) leads to a substantial drop across all tasks, as the policy no longer receives a stable, trajectory-wide signal to encourage long-range coherence. Similarly, discarding the step relative advantages (GiGPO$_{\text{w/o } A^S}$) results in pronounced declines, particularly on more complex and demanding tasks such as Cool, Pick2, and WebShop, which may require nuanced training feedback at each decision step. In such cases, precise per-step credit assignment is essential for effective learning and policy refinement. Moreover, we can see that the relative performance gap between GiGPO$_{\text{w/ std}}$ and GiGPO$_{\text{w/o std}}$ is comparatively minor compared to that observed in structural ablations. This suggests that the combination of episode- and step-level signals is the primary driver of performance gains, and that each component is crucial for training LLM agents effectively.

5.4 Dynamics of Step-Level Group

In this part, we examine how the distribution of step-level groups evolves throughout training to better understand the utility of GiGPO. We use Qwen2.5-1.5B-Instruct as the base model. We train the LLM agent in ALFWorld and track changes in step-level group sizes throughout training.

As shown in Figure 5, we observe that step-level groups of size 1 (i.e., those with $|G^S(\tilde{\boldsymbol{s}})| = 1$) only account for < 35% throughout training. This indicates that the majority of states (over 65%) recur across trajectories and therefore contribute to achor state grouping. Moreover, at iteration 10, large group sizes $|G^S(\tilde{\boldsymbol{s}})| \geq 10$ account for over 20%, reflecting behavioral redundancy in the early stages of training. This is consistent with the fact that immature policies often produce invalid actions or fall into repetitive loops. As training progresses, the group size distribution changes markedly. By iteration 75, we observe a substantial reduction in extreme group sizes: $10 \leq |G^S(\tilde{\boldsymbol{s}})| < 50$ drops from 16.2% to 12.1% and $|G^S(\tilde{\boldsymbol{s}})| \geq 50$ drops from 5.6% to 3.1%. This shift suggests that the agent is learning to avoid previously common dead ends and invalid actions, and is beginning to exhibit more diverse and purposeful decision-making. At iteration 140, the distribution becomes tightly

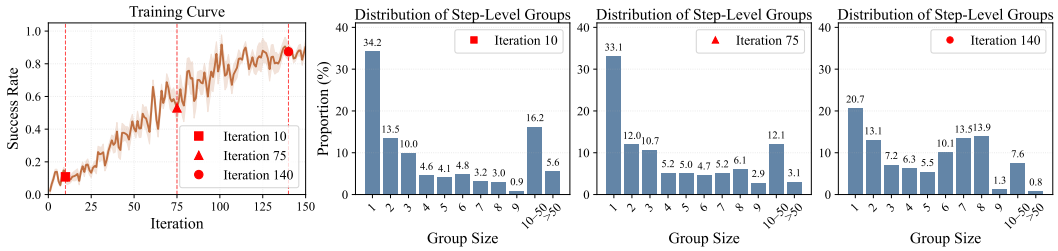

Figure 5: Dynamics of step-level groups during the training in ALFWorld. **Left**: Success rate over training iterations. Vertical red markers denote selected checkpoints (Iterations 10, 75, 140). **Right**: Distribution of step-level group sizes at those checkpoints. For each anchor state $\tilde{s} \in \mathcal{U}$, the group size is given by $|G^S(\tilde{s})|$.

concentrated around group sizes of 6 to 8. Given that N is set to 8, this convergence implies that all 8 trajectories within an episode group are now behaving consistently: LLM agent has learned a coherent and robust policy for completing the task, aligning with the plateau in success rate ($> 80\%$).

5.5 Computational Budget

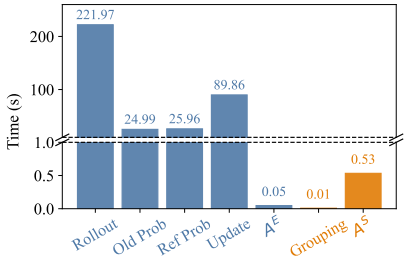

Figure 6: Per-iteration training time breakdown of GiGPO. Blue bars indicate shared components with GRPO. Orange bars show GiGPO-specific additions. The y-axis uses a broken scale to accommodate small values.

Lastly, we analyze the computational budget of GiGPO. GiGPO shares the same core architecture as GRPO, including multi-turn rollouts, computation of old and reference probabilities, and clipped policy updates. Both approaches are critic-free and operate with a single actor LLM, thereby resulting in identical GPU memory usage and LLM rollout costs. The primary additions introduced by GiGPO are the step-relative advantage estimation components, as described in Section 4.2. To evaluate their costs, we train an LLM agent in ALFWorld using Qwen2.5-1.5B-Instruct and record a detailed breakdown of per-iteration training time.

As shown in Figure 6, the additional components incur little to no additional time cost compared to dominant operations such as rollouts, computation of old and reference probabilities, and policy updates, whose total time cost reaches 362.83s per iteration. In contrast, anchor state grouping (involving hashmap lookups) takes only 0.01s per iteration, and the step-relative advantage computation (involving simple arithmetic) adds just 0.53s. These operations account for $< 0.002\%$ of the total per-iteration training time, demonstrating that GiGPO shares the same high computational efficiency as GRPO.

6 Conclusions and Limitations

In this work, we proposed GiGPO, a novel group-based RL algorithm to tackle the credit assignment challenge in long-horizon LLM agent training. GiGPO introduces a hierarchical advantage estimation that enables fine-grained per-step credit assignment while retaining the efficiency and stability of group-based RL. By retroactively grouping steps that share the same state across trajectories, it achieves this without incurring additional LLM rollout or GPU memory overhead. Empirical evaluations across two complex environments, ALFWorld and WebShop, demonstrate that GiGPO significantly outperforms both prompt-based agents and prior RL methods. A potential limitation of GiGPO is its reliance on state matching for anchor group construction. In highly complex environments, identical states may be hard to detect due to noise or subtle differences. Despite this, GiGPO still retains a strong performance lower bound: in the extreme case where no states are repeated across trajectories (i.e., $A^S = 0$), it naturally degrades to GRPO, preserving GRPO's effectiveness and stability in credit assignment. However, we suggest a more robust solution: introducing state similarity through embeddings or approximate matching, which could better capture structurally equivalent states. We leave this exploration as a promising direction for future work.